



Status and plans of the submission on GPUs

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On Behalf of the CMS Submission Infrastructure team

Acknowledgements: Alan Malta

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Outline

- Discuss how GPUs have been integrated
- Present results of the integration
- Bring up open questions and evolution proposals

Integration of GPUs in CMS

- Enable CMS WM and SI to make use of **distributed GPU** resources
- **“Multidisciplinary” work** with multiple groups and multiple pieces needing to work together
 - CMSSW
 - Ability to offload to GPUs
 - Workflow configuration
 - Workload Management
 - ReqMgr and WMAgent GPU support
 - Global pool
 - Resource addition
 - Pool configuration
 - McM changes
- **Many areas** involved (now or in the future!)
- Efforts kickstarted by the Fall O&C week 2019
 - Lot of effort in cross group coordination, workshops, etc

Core Software S. Muzaffar, M. Kortelainen	Computing Operations C. Paus, N. Smith	Dyn. Res. Provisioning D. Spiga, C. Wissing	Facility Services G. Bagliesi, S. Lammel
Simulation V. Ivantchenko, S. Bein	Workload/Data Mgt. Devel. K. Lannon, K. Ellis	Reconstruction S. Krutelyov, J. Pata	Resource Management J. Flix, D. Lange
Monitoring & Analytics B. K. Jashal, F. Legger	Offline Release Planning A. Perrotta, Q. Li	Upgrade Software P. Srimanobhas, A. Di Florio	Submission Infra. A. Perez-Calero, M. Mascheroni
Analysis Infra. & Support S. Belforte, S. Malik, M. Tonjes	Upgrade R&D and TDR M. Girone, D. Elvira	Web Services & Security A. Pfeiffer, P. Paparrigopoulos	
Generators * S. Bhattacharya, G. Chahal	Machine Learning * J.R. Vlimant, G. Kasieczka	L1 Software ** V. Rekovic, C. Caillol	DPOA *** K. Lassila-Perini, C. Lange (Dep.)

GPU workflows in CMSSW

- Main switch **RequiresGPU:**

- True
- False
- (Optional*)

- GPU Parameters

runTheMatrix

```
{'AcquisitionEra': 'CMSSW_12_1_0_pre3',
'CMSSWVersion': 'CMSSW_12_1_0_pre3',
'Campaign': 'CMSSW_12_1_0_pre3',
'ConfigCacheUrl': 'https://cmsweb-testbed.cern.ch/couchdb',
'DQMConfigCacheID': '5c411491cd1ae3438c5ffb301612591e',
'DQMUploadUrl': 'https://cmsweb.cern.ch/dqm/relval',
'DbsUrl':
'https://cmsweb-testbed.cern.ch/dbs/int/global/DBSReader',
'EnableHarvesting': 'True',
'GlobalTag': '121X_mcRun3_2021_realistic_v2',
'Group': 'ppd',
'Memory': 3000,
'Multicore': 1,
'PrepID': 'CMSSW_12_1_0_pre3_1633006291-TTbar_14TeV',
'ProcessingString': '121X_mcRun3_2021_realistic_v2',
'ProcessingVersion': 1,
'RequestPriority': 500000,
'RequestString': 'RVCMSW_12_1_0_pre3TTbar_14TeV',
'RequestType': 'TaskChain',
'Requestor': 'srimanob',
'ScamArch': 'slc7_amd64_gcc900',
'SizePerEvent': 1234,
'SubRequestType': 'RelVal',
'Task1': {'AcquisitionEra': 'CMSSW_12_1_0_pre3',
'ConfigCacheID': '5c411491cd1ae3438c5ffb3016125ba7',
'EventStreams': 0,
'EventsPerJob': 100,
'EventsPerLumi': 100,
'GlobalTag': '121X_mcRun3_2021_realistic_v2',
'KeepOutput': True,
'Memory': 13500,
'Multicore': 8,
'PrimaryDataset': 'RelValTTbar_14TeV',
'ProcessingString': '121X_mcRun3_2021_realistic_v2',
'RequestNumEvents': 9000,
'Seeding': 'AutomaticSeeding',
'SplittingAlgo': 'EventBased',
'TaskName': 'TTbar_14TeV_TuneCP5_2021_GenSim'},
```

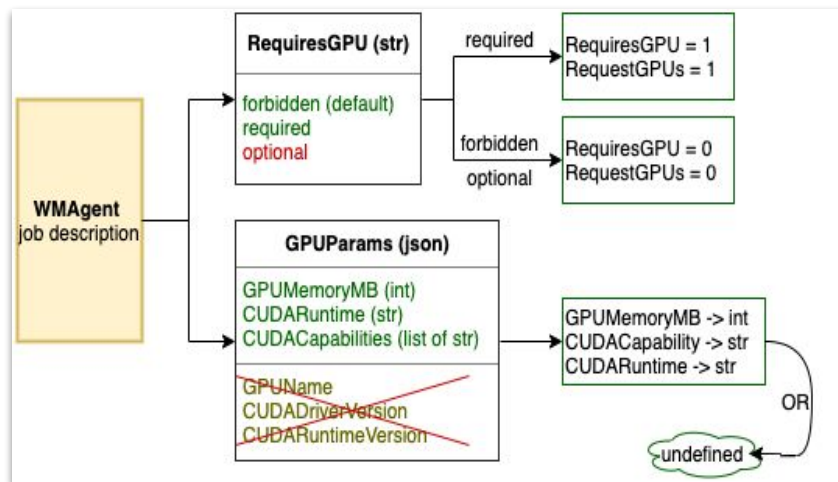
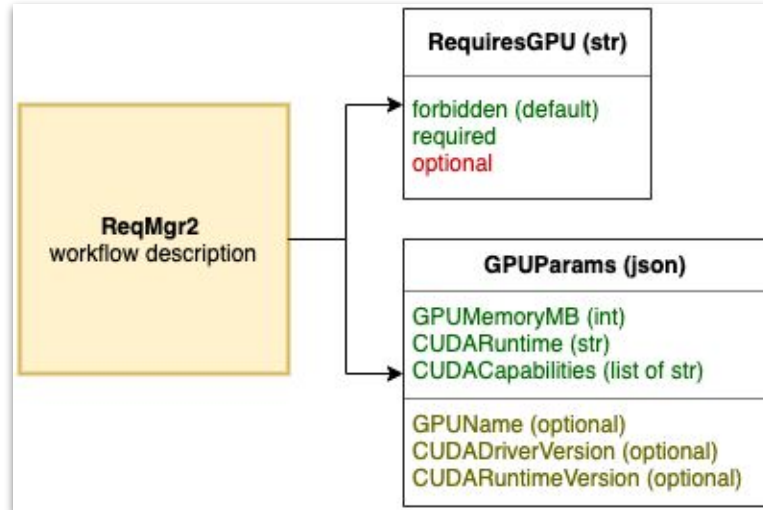
```
'Task2': {'AcquisitionEra': 'CMSSW_12_1_0_pre3',
'ConfigCacheID': '5c411491cd1ae3438c5ffb3016126fd0',
'EventStreams': 0,
'GlobalTag': '121X_mcRun3_2021_realistic_v2',
'InputFromOutputModule': 'FEVTDEBUGOutput',
'InputTask': 'TTbar_14TeV_TuneCP5_2021_GenSim',
'KeepOutput': True,
'LumisPerJob': 10,
'Memory': 13500,
'Multicore': 8,
'ProcessingString': '121X_mcRun3_2021_realistic_v2',
'SplittingAlgo': 'LumiBased',
'TaskName': 'Digi_2021'},
'Task3': {'AcquisitionEra': 'CMSSW_12_1_0_pre3',
'ConfigCacheID': '5c411491cd1ae3438c5ffb3016128a40',
'EventStreams': 0,
'GPUParams': {'GPUMemoryMB': 8000,
'CUADACapabilities': ["6.0", "6.1",
"6.2", "7.0", "7.2", "7.5", "8.0", "8.6"],
'CUADARuntime': "11.4"},
'GlobalTag': '121X_mcRun3_2021_realistic_v2',
'InputFromOutputModule': 'FEVTDEBUGHLToutput',
'InputTask': 'Digi_2021',
'KeepOutput': True,
'LumisPerJob': 10,
'Memory': 13500,
'Multicore': 8,
'ProcessingString': '121X_mcRun3_2021_realistic_v2',
'RequiresGPU': 'required',
'SplittingAlgo': 'LumiBased',
'TaskName':
'Reco_Patatrack_PixelOnlyTripletsGPU_2021'},
'TaskChain': 3,
'TimePerEvent': 10}
```

Workflow configuration

Workload Management

GPU support within the WM system

- There are **two layers within the WM** system supporting GPU workload.
 - ReqMgr2: with the workflow description (only ReReco and TaskChain at the moment)
 - WMAgent: with the **job creation** and definition
- Workflows that desire to use GPU resources need to define two new key/value pairs
 - RequiresGPU**
 - GPUParams**
- Gets described to **a total of 5 htcondor job classads**, defining whether GPUs must be used or not and which settings to use.



From Workload Management to Global pool

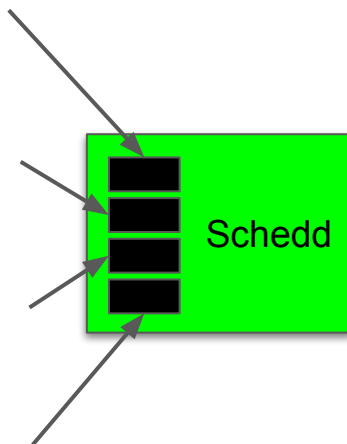
```
[mmascher@lxplus737 ~]$ condor_history -const 'WMAgent_ReqName ==
"srimanob_RVCMSSW_12_1_0_pre3TTbar_14TeV__PHATTEST2_210930_184013_2586" && requestgpus==1 && JobStatus=?=4' -pool vocms0803 -name vocms0262
-1 | egrep '^ (CRequiresGPU|RequestGPUs|GPUMemoryMB|CUDA_Runtime|CUDA_Capability) '
```

```
CUDACapability = "6.0,6.1,6.2,7.0,7.2,7.5,8.0,8.6"
CUDA_Runtime = "11.4"
GPUMemoryMB = 8000
RequestGPUs = 1
RequiresGPU = 1
```

```
CUDACapability = "6.0,6.1,6.2,7.0,7.2,7.5,8.0,8.6"
CUDA_Runtime = "11.4"
GPUMemoryMB = 8000
RequestGPUs = 1
RequiresGPU = 1
```

```
CUDACapability = "6.0,6.1,6.2,7.0,7.2,7.5,8.0,8.6"
CUDA_Runtime = "11.4"
GPUMemoryMB = 8000
RequestGPUs = 1
RequiresGPU = 1
```

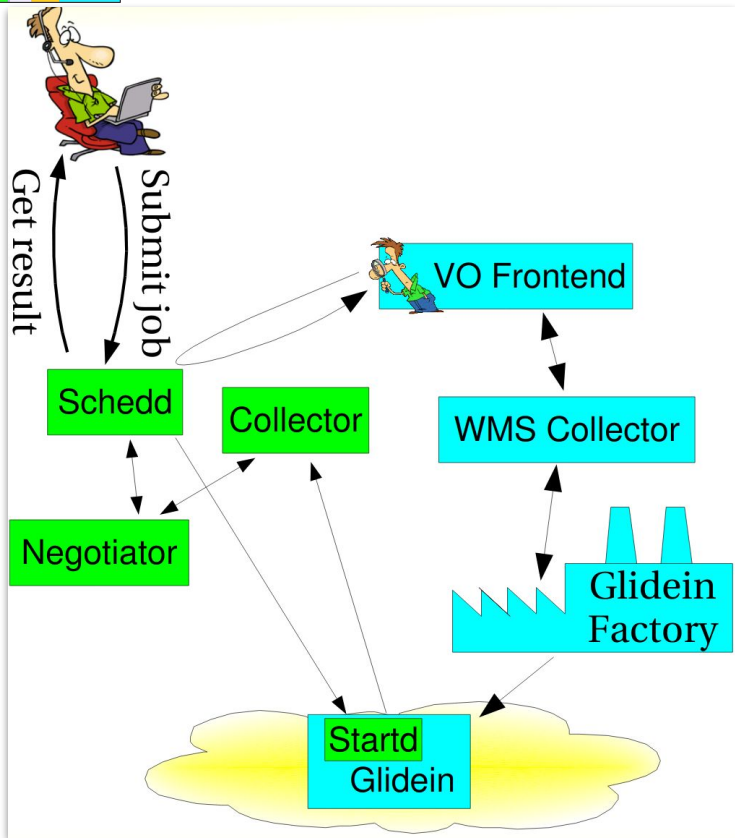
```
CUDACapability = "6.0,6.1,6.2,7.0,7.2,7.5,8.0,8.6"
CUDA_Runtime = "11.4"
GPUMemoryMB = 8000
RequestGPUs = 1
RequiresGPU = 1
```



- Jobs are in the schedd
- Need to **find resources** to run!
- Resources are **provided by GlideinWMS**

Submission Infrastructure

GPUs in a Dynamic HTCondor pool



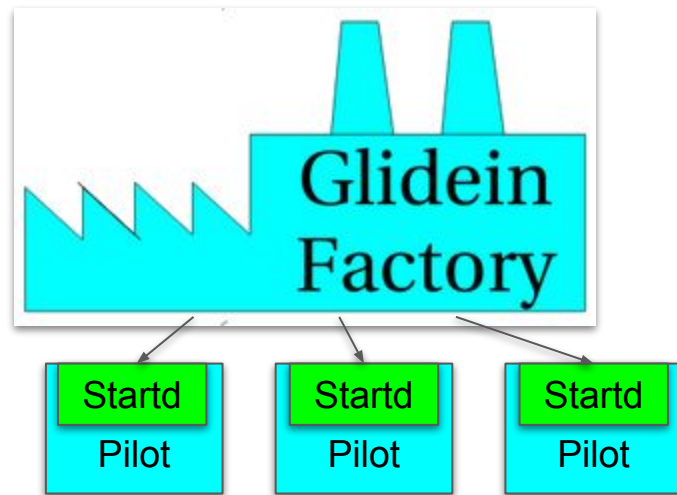
Two matchmaking stages in Submission Infrastructure:
resource allocation (GlideinWMS) and job to slot
matchmaking (HTCondor)

1. GlideinWMS and pilot jobs
 - Get the resources from the sites
2. HTCondor matchmaking
 - Attach resources to jobs

Resource Allocation and First Matchmaking

- **First matchmaking stage: Jobs with `RequiresGPU = True` trigger pilot job submission to sites with GPUs**
 - No other GPU parameters are checked here
- **Resource route statically configured: pilot factory entries**
 - No GPU pledges => No VO Card
 - Resource attributes agreed upon with the site admins
 - GPU slot lifetime (24h? 48h?)
 - Typically slots are configured with 8 CPU cores, 1GPU, 2.5 Gb/core RAM

```
<entry name="CMSHTPC_T2_US_Wisconsin_cmsggrid01_gpu" auth_method="grid_proxy"
gatekeeper="cmsggrid01.hep.wisc.edu cmsggrid01.hep.wisc.edu:9619" gridtype="condor" ...>
  <config>
    ... Parameters related to limits and submit rates ...
    <submit_attrs>
      <submit_attr name="maxMemory" value="20000"/>
      <submit_attr name="xcount" value="8"/>
      <submit_attr name="Request_GPUs" value="1"/>
    </submit_attrs>
  </config>
  <attrs>
    <attr name="GLIDEIN_CMSSite" ... type="string" value="T2_US_Wisconsin"/>
    <attr name="GLIDEIN_CPUS" ... type="string" value="8"/>
    <attr name="GLIDEIN_MaxMemMBs" ... type="int" value="20240"/>
    <attr name="GLIDEIN_Max_Walltime" ... type="int" value="216000"/>
    <attr name="GLIDEIN_Resource_Slots" ... type="string" value="GPUs,1,type=main"/>
    <attr name="GLIDEIN_Site" ... type="string" value="Wisconsin"/>
    <attr name="GLIDEIN_Supported_VOs" ... type="string" value="CMSGPU"/>
  </attrs>
</entry>
```



Second matchmaking matches job attributes/requirements with the machine ones

- **GPU properties discovered** when pilot jobs starts
 - `condor_gpu_discovery` used for most of the matchmaking attributes
 - `CMS_CUDA_SUPPORTED_RUNTIMES` from CMS script in `cvmfs`
- A properly configured **startd joins the Global Pool**
 - a slot to match GPU jobs
- **WMAgent jobs can start** if a match is found!

Job attributes

```
CUDACapability =
"6.0,6.1,6.2,7.0,7.2,7.5,8.0,8.6"
CUDARuntime = "11.4"
GPUMemoryMB = 8000
RequestGPUs = 1
RequiresGPU = 1
...
```

Collector + Negotiator

```
...
Machine.CUDACapability in Job.CUDACapability
Job.CUDARuntime in Machine.CMS_CUDA_SUPPORTED_RUNTIMES
Job.GPUs <= Machine.GPUs
...
```

Machine attributes

```
CPUs = 8
GPUs = 2
CUDACapability = 8.0
CUDAClockMhz = 1410.0
CUDAComputeUnits = 108
CUDACoresPerCU = 64
CUDADeviceName = "NVIDIA A100-PCIE-40GB"
CUDADriverVersion = 11.3
CUDAECCEEnabled = true
CUDAGlobalMemoryMb = 40536
CUDAMaxSupportedVersion = 11030
CMS_CUDA_SUPPORTED_RUNTIMES =
10.1,10.2,11.0,11.1,...
...
```

Results

Monitoring GPUs available to CMS

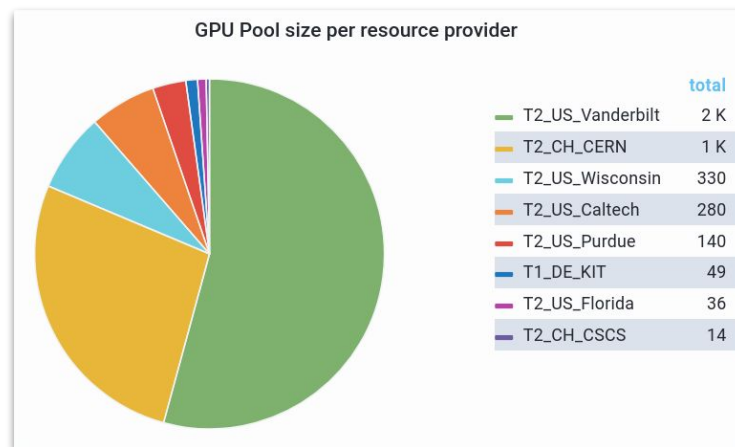
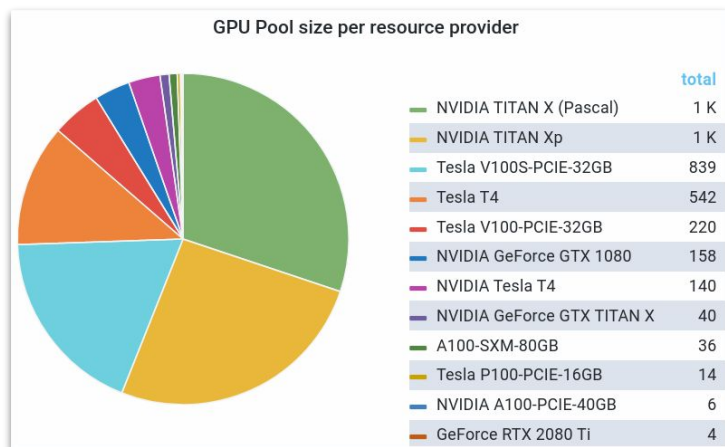
- **Regularly scanning** the Global Pool for **GPU resources** and their properties
 - Submit pilots that run the **HTCondor GPU discovery tool to fill the table**
- Resources are opportunistic in nature for now, sites voluntarily adding GPUs to CMS pool
 - So far: T1_DE_KIT, T2_US_Florida, T2_US_Caltech, T2_US_MIT, T2_US_Purdue, T2_US_Wisconsin, T2_US_Vanderbilt, T2_CH_CERN, T2_CH_CSCS, T3_US_NotreDame, T3_US_OSG

Site	Entry	CUDACapability	CUDAClockMhz	CUDAComputeUnits	CUDACoresPerCU	CUDADeviceName	CUDADriverVersion	CUDAECCEEnabled	CUDAGlobalMemoryMb
T2_US_Caltech	CMSHTPC_T2_US_Caltech_cit2_gpu	5.2	1216	24	128	GeForce GTX TITAN X	11	false	12213
T2_US_Caltech	CMSHTPC_T2_US_Caltech_cit_gpu	5.2	1216	24	128	GeForce GTX TITAN X	11	false	12213
T2_US_Caltech	CMSHTPC_T2_US_Caltech_cit3_gpu	5.2	1216	24	128	GeForce GTX TITAN X	11	false	12213
T3_US_OSG	HCC_US_Omaha_crane_gpu	6.0	1329	56	64	Tesla P100-PCIE-16GB	11	true	16281
T3_US_OSG	HCC_US_Omaha_crane_gpu	6.0	1329	56	64	Tesla P100-PCIE-12GB	11	true	12198
T3_US_OSG	HCC_US_Omaha_crane_gpu	6.1	1582	28	128	GeForce GTX 1080 Ti	11	false	11178
T3_US_OSG	HCC_US_Omaha_crane_gpu	6.1	1709	10	128	GeForce GTX 1060 6...	11	false	6078
T2_US_Vanderbilt	CMS_T2_US_Vanderbilt_ce6_gpu	6.1	1582	30	128	TITAN Xp	10	false	12196
T2_US_Vanderbilt	CMS_T2_US_Vanderbilt_ce5_gpu	6.1	1582	30	128	TITAN Xp	10	false	12196
T2_US_Vanderbilt	CMS_T2_US_Vanderbilt_ce6_gpu	6.1	1531	28	128	TITAN X (Pascal)	10	false	12196
T2_US_Vanderbilt	CMS_T2_US_Vanderbilt_ce5_gpu	6.1	1531	28	128	TITAN X (Pascal)	10	false	12196
T3_US_OSG	HCC_US_Omaha_crane_gpu	7.0	1380	80	64	Tesla V100-PCIE-32GB	11	true	32510
T2_US_Purdue	CMSHTPC_T2_US_Purdue_Hamme...	7.5	1590	40	64	Tesla T4	11	true	15110
T3_US_OSG	HCC_US_Omaha_crane_gpu	7.5	1815	48	64	Quadro RTX 5000	11	false	16125
T3_US_OSG	Glow_US_Syracuse2_condor_gpu	7.5	1620	72	64	Quadro RTX 6000	11	true	22699
T3_US_OSG	HCC_US_Omaha_crane_gpu	7.5	1770	72	64	Quadro RTX 8000	11	false	48601
T3_US_OSG	Glow_US_Syracuse3_condor_gpu	7.5	1815	48	64	Quadro RTX 5000	11	false	16125
T3_US_OSG	Glow_US_Syracuse2_condor_gpu	7.5	1815	48	64	Quadro RTX 5000	11	false	16125
T2_US_Wisconsin	CMSHTPC_T2_US_Wisconsin_cms...	7.5	1590	40	64	Tesla T4	11	true	15110
T2_US_Wisconsin	CMSHTPC_T2_US_Wisconsin_cms...	7.5	1590	40	64	Tesla T4	11	true	15110
T2_US_Wisconsin	CMSHTPC_T2_US_Wisconsin_cms...	7.5	1590	40	64	Tesla T4	11	true	15110

GPU Integration Status

- Successfully executed tests with real workflows in ITB!
- Exercised a variety of GPUs!

Number of recent jobs run per GPU type and provider



Future work

Use GPU “optional”: SI proposal

Profit from framework flexibility to dynamically adapt workload to CPU or CPU+GPU environment

Two matchmaking stages where policies are needed, implemented on the basis of job attributes

SI + WM **proposal** is as follows

- RequiresGPU: The workflow MUST use GPUs, thus defines whether or not GPU pilots will be requested by the GlideinWMS FE (GlideinWMS matchmaking)
- RequestGPUs: GPU resources to be matched in the second (HTCondor) matchmaking

Cases:

- RequiresGPU = True && RequestGPUs>0, will trigger GPU pilot, then match the slot
- RequiresGPU = False && RequestGPUs>0, will not trigger GPU pilots but CAN match already existing GPU slots.
 - Preference for CPU+GPU over CPU slots? Use “RequiresGPU = optional”
- RequiresGPU = False && RequestGPUs=0, purely CPU task

Further Challenges

- Come up with a policy for GPUs in StepChain workflows
 - With multi-step jobs (multiple cmsRun with different settings). Not all of them capable of making use of the GPU.
- Making the most effective use of the GPUs
 - Oversubscribe GPUs? Technical solutions available in HTCondor
- Need to identify the T0 use cases and integrate this feature to T0 workflows.

Conclusions and Plans

- Despite being **opportunistic**, already **GPUs are available** in the Global Pool from a number of sites
- Demonstrated ability to use them with WMAgent workflows
 - TaskChain release validation test workflow executed in ITB
 - ReReco development tests only
- Possibility to request GPU **ReReco and TaskChain workflows** now available
- **Challenges** ahead
 - Find way to run **multi-steps jobs** without sacrificing GPU efficiency
 - Enable **GPU=optional** (start with the simple proposal and evolve from that?)
 - **Validation** of sites with GPUs (especially relevant once sites start to pledge GPUs)
 - Benchmarking and accounting

Backup Slides

GPU benchmarking & accounting

- Why **benchmarking**?
 - Pledges definition and resource acquisition
 - A critical element in proper **accounting of resource utilization**
- The **current standard** HEP benchmark: **HS06** (based on 2006 specification, **no GPU**)
 - **HEPScore** as a replacement that **has been developed** by the HEPiX Benchmarking Working Group ([source](#))
 - Inclusion of **GPUs** is an **ongoing development effort** (using event processing throughput as performance metric)
- While the proper procedure for GPU usage accounting is defined, our simple proposal to start accounting GPU usage would be to record all the data related to job execution time, along with GPU model, CUDA libraries, etc.
 - Use “GPU time”, as provided by HTCondor, in addition to `GPUsAverageUsage` and `GPUsMemoryUsage`
- A **GPU benchmark** will be needed to reasonably predict **scaled execution time of CMS workloads**
 - A key parameter for an efficient matchmaking of jobs to slots
 - Up to now, reasonable assumption of approx. equal run times on all CPU resources
 - The use of **GPU** coprocessors introduce a high degree of heterogeneity in the resources we can access, and their relative throughputs
 - Observed a 10x execution time difference on our very simple test jobs!

Refs for WM:

<https://github.com/dmwm/WMCore/wiki/GPU-Support#gpu-job-description-in-wmcore-x-glideinwms>

<https://github.com/dmwm/WMCore/issues/10393>

Previous slides:

https://indico.cern.ch/event/842166/contributions/3581503/attachments/1930275/3196841/20191021_GPU_usage_WM_SI_Ops.pdf

<https://agenda.hep.wisc.edu/event/1579/contributions/23061/attachments/7926/8974/Evolution%20of%20the%20CMS%20Submission%20Infrastructure%20to%20Support%20Heterogeneous%20Resources.pdf>